

Package ‘gwk’

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Type Package

Title Convenience Wrappers and Workflows for Geographically Weighted Analysis

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Description A toolkit of convenience wrappers and reproducible workflows around 'GWmodel' and related spatial packages ('sf', 'terra', 'sp') for geographically weighted analysis. Provides point-, polygon-, and county-level geographically weighted summary-statistics pipelines, distance-metric presets, and utilities to reduce results estimated across a grid of kernel and distance-metric settings to a single per-unit consensus, with optional Queen-contiguity reconciliation. The package wraps, and does not replace, the underlying estimators.

License GPL-3

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knitr,
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Remotes github::UrbanInstitute/urbnmapr

URL <https://github.com/ftsiboe/gwkit>

BugReports <https://github.com/ftsiboe/gwkit/issues>

Config/testthat/edition 3

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estimate_gwlag	<i>Geographically weighted spatial lag (neighbour-weighted mean)</i>
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Description

For each target unit, computes the geographically weighted mean of one or more value columns over the source units, using `GWmodel::gw.weight()` weights on a `GWmodel::gw.dist()` distance matrix. With `include_self = FALSE` the focal unit is excluded from its own neighbourhood, yielding a strict neighbour mean (a spatial lag). Point vs polygon mode is class-detected: pass an `sf`/`SpatVector` (as data, or via geometry for panels) for polygon units, each reduced to its point-on-surface; otherwise the `coords` columns of data are used.

Usage

```
estimate_gwlag(  
  data,  
  unit,  
  value_cols,  
  geometry = NULL,  
  coords = c("longitude", "latitude"),  
  poly_id = unit,  
  predict = NULL,  
)
```

```

distance_metric = "Euclidean",
kernel = "bisquare",
adaptive = FALSE,
bw = NULL,
bw_response = NULL,
bw_approach = "CV",
include_self = FALSE
)

```

Arguments

data	A data.table/data frame of the SOURCE units (one row per unit) with unit, all value_cols, and - in point mode - the coords columns; or an sf/SpatVector carrying both attributes and geometry (polygon mode).
unit	Character; the unit-id column.
value_cols	Character vector of numeric columns to lag. Each returns a column named <value_col>_LM.
geometry	Optional sf/SpatVector polygon layer (id field poly_id matching unit) supplying per-unit geometry. If NULL and data is not spatial, point mode uses coords.
coords	Length-2 character vector naming longitude/latitude columns (point mode). Default c("longitude", "latitude").
poly_id	Name of the id field in the polygon layer. Default: unit.
predict	Optional target restriction: a data frame of point targets (point mode) or a character id vector / data frame with unit (polygon mode). Defaults to the units in data.
distance_metric	One of gw_distance_metric_names(). Default "Euclidean".
kernel	GW kernel. Default "bisquare".
adaptive	Logical; adaptive (kNN) bandwidth if TRUE. Default FALSE (fixed-distance); note estimate_gwr()/estimate_gwss() default to TRUE.
bw	Optional pre-computed bandwidth. If NULL, selected once via GWmodel::bw.gwr(bw_response ~ 1) on the source set.
bw_response	Column used for bandwidth selection when bw is NULL. Default: the first value_cols.
bw_approach	Bandwidth criterion. Default "CV".
include_self	Logical; keep the focal unit in its own neighbourhood. Default FALSE (strict neighbour mean).

Value

A data.table with one row per target unit: the unit id and one <value_col>_LM column per value column. The selected bandwidth is attached as attr(, "bandwidth").

See Also

Other Geographically weighted summaries: [estimate_gwss\(\)](#)

estimate_gwr

*Geographically weighted local regression***Description**

One geographically weighted regression covering plain GWR, fixed-effects GWR, and the mean/variance decomposition, over the shared local WLS engine. Point vs polygon mode is detected by class: pass an sf/terra SpatVector (as data, or via geometry for panel inputs) for polygon units - each reduced to its point-on-surface; otherwise the coords columns of data are used.

Usage

```
estimate_gwr(
  data,
  unit,
  formula,
  geometry = NULL,
  coords = c("longitude", "latitude"),
  poly_id = unit,
  predict = NULL,
  panel = NULL,
  time = NULL,
  distance_metric = "Euclidean",
  kernel = "bisquare",
  adaptive = TRUE,
  bw = NULL,
  bw_approach = "CV",
  variance = FALSE,
  fit_stats = FALSE,
  standard_errors = FALSE,
  terms = NULL,
  eps = 0.00000001
)
```

Arguments

data	A data.table/data frame with the model variables (and, for fixed effects, the panel/time columns), OR an sf/SpatVector carrying both attributes and geometry (polygon mode).
unit	Character; the spatial-unit id column (equal to panel in fixed-effects mode).
formula	A model formula, e.g. $y \sim x$ or $y \sim x1 + x2$.
geometry	Optional sf/SpatVector polygon layer (whose poly_id matches unit) supplying per-unit geometry when data is a plain (e.g. panel) table. If NULL and data is not spatial, point mode is used.
coords	Length-2 character vector naming longitude/latitude columns (point mode). Default c("longitude", "latitude").
poly_id	Name of the id field in the polygon layer. Default: unit.
predict	Optional target restriction: a data frame of point targets (point mode) or a character id vector / data frame with unit (polygon mode). Defaults to the unique units in data.

panel, time	Optional fixed-effects panel id and time id. If panel is supplied the panel fixed effects are absorbed (within transform) before the GW fit and model_estimator is "gwfe"; otherwise "gwr".
distance_metric	One of gw_distance_metric_names(). Default "Euclidean".
kernel	GW kernel. Default "bisquare".
adaptive	Logical; adaptive (kNN) bandwidth if TRUE. Default TRUE.
bw	Optional pre-computed bandwidth. If NULL, selected once via GWmodel::bw.gwr() on the (demeaned) response, intercept-only (. ~ 1); the bandwidth targets the response surface, not the full covariate model. Model columns may be named anything (including X/Y) - internal coordinate columns are dot-prefixed and will not collide.
bw_approach	Bandwidth criterion. Default "CV".
variance	Logical; also fit the local slope of the log-squared residuals (a heteroskedasticity companion), returned as estimand == "variance" rows. Requires the targets to cover the source units. Default FALSE.
fit_stats	Logical; attach local within-r_squared and n_obs. Default FALSE.
standard_errors	Logical; also return analytic WLS se/tv/pv. Default FALSE - coefficients only (identical to lm); use the bootstrap.
terms	Optional character vector restricting the returned terms.
eps	Added to squared residuals before the log (variance mode). Default 1e-8.

Value

A data.table, one row per unit_id x term x estimand (estimand in "mean"/"variance"): unit_id, term, estimand, unit_level, estimate, se, tv, pv, (when fit_stats) r_squared, n_obs, and model_estimator. Bandwidth in attr(, "bandwidth"); number of panels absorbed (FE mode) in attr(, "NFE").

estimate_gwss	<i>Estimate geographically weighted summary statistics (GWSS)</i>
---------------	---

Description

One entry point over GWmodel::gwss(): at each prediction location it returns local summary statistics (local mean, SD, CV, skewness, and local covariance / correlation between variables). Point vs polygon mode is detected by class.

Usage

```
estimate_gwss(
  data,
  variable_list,
  geometry = NULL,
  us_counties = FALSE,
  id_col = NULL,
  coords = c("longitude", "latitude"),
```

```

predict = NULL,
distance_metric = "Euclidean",
kernel = "gaussian",
adaptive = TRUE,
approach = "CV",
target_crs = 5070,
draw_rate = 0.5,
feasible_radius = 100,
identifiers = NULL,
state_fips_limits = NULL
)

```

Arguments

<code>data</code>	A <code>data.table</code> /data frame with <code>variable_list</code> and either the <code>coords</code> columns (point mode) or the <code>id_col</code> join key (polygon mode).
<code>variable_list</code>	Character vector of numeric variable column(s) to summarize (passed to <code>GWmodel::gwss(vars =)</code>). The bandwidth uses the first.
<code>geometry</code>	Optional <code>sf</code> / <code>SpatVector</code> polygon layer whose <code>id_col</code> field matches <code>data[[id_col]]</code> (polygon mode). If <code>NULL</code> and <code>us_counties = FALSE</code> , point mode is used.
<code>us_counties</code>	Logical; if <code>TRUE</code> , pull the U.S. county polygons from <code>urbnmapr</code> (polygon mode). <code>id_col</code> defaults to <code>"county_fips"</code> .
<code>id_col</code>	Character; the unit id column shared by <code>data</code> and <code>geometry</code> (polygon mode). Required in polygon mode.
<code>coords</code>	Length-2 character vector naming longitude/latitude columns (point mode). Default <code>c("longitude", "latitude")</code> .
<code>predict</code>	Optional prediction locations (point mode): a data frame with the <code>coords</code> columns (and any identifiers). Defaults to the observed points.
<code>distance_metric</code>	One of <code>gw_distance_metric_names()</code> . Default <code>"Euclidean"</code> .
<code>kernel</code>	GW kernel. Default <code>"gaussian"</code> .
<code>adaptive</code>	Logical; adaptive (kNN) bandwidth if <code>TRUE</code> . Default <code>TRUE</code> .
<code>approach</code>	Bandwidth criterion, one of <code>"CV"</code> , <code>"AIC"</code> , <code>"AICc"</code> . Default <code>"CV"</code> .
<code>target_crs</code>	Integer EPSG code (meters) for projection (polygon geometry and the point-mode radius screen). Default 5070 (NAD83 / CONUS Albers).
<code>draw_rate</code>	Fraction of observed units used for bandwidth selection. Default 0.5.
<code>feasible_radius</code>	Radius in miles for pre-screening prediction points (point mode only). Default 100.
<code>identifiers</code>	Optional character vector of columns in <code>predict</code> (point mode) to carry through to the output.
<code>state_fips_limits</code>	Optional state FIPS to restrict the county set (<code>us_counties</code> /polygon mode, when a <code>state_fips</code> column is present).

Value

A `data.table` of GWSS statistics with one row per prediction unit, the unit id (polygon mode) or coordinates + identifiers (point mode). The bandwidth and distance parameters are attached as attributes.

See Also

Other Geographically weighted summaries: [estimate_gwlag\(\)](#)

fixed_effect_model_data_prep

Prepare and demean data for fixed-effects models

Description

This function

1. Filters to complete cases on the specified panel, time, weight, variables, and output
2. If output is NULL, creates a dummy output column filled with 1s
3. Drops any panel with only one observation
4. Computes within-panel means for the output + each variable in varlist (`_mean_i`)
5. Computes overall sample means for the same set of variables (`_mean`)
6. Replaces each variable in varlist by `value - within_panel_mean + overall_mean`

Usage

```
fixed_effect_model_data_prep(
  data,
  varlist,
  panel,
  time,
  wvar = NULL,
  output = NULL
)
```

Arguments

<code>data</code>	A <code>data.frame</code> or <code>data.table</code> containing the data.
<code>varlist</code>	Character vector of variable names to be demeaned.
<code>panel</code>	Character vector of column name(s) defining the panel identifier.
<code>time</code>	Character scalar name of the time variable.
<code>wvar</code>	Character scalar name of a variable to keep but <i>not</i> demean (optional, default NULL).
<code>output</code>	Character scalar name of an output variable whose means are computed but not altered; if NULL, a dummy column named "output" is created (optional, default NULL).

Value

A list with components

- **data**: a `data.table` containing
 - the original panel, time, wvar, varlist, and output columns
 - two mean columns for each of `c(output, varlist)`: `<name>_mean_i` (within-panel) and `<name>_mean` (overall)
- **NFE**: the number of panels with more than one observation

gw_consensus_class	<i>Consensus class across the kernel x distance-metric domain</i>
--------------------	---

Description

Collapses a stacked table of per-unit classifications (one row per unit per distance_metric x kernel setting, within a history window) to a single consensus class per unit: the mode across settings (modal_class) plus a first-order Queen contiguity vote for optional spatial de-speckling (queen_class), with ties resolved by widening the Queen order. Point vs polygon mode is class-detected: pass an sf/SpatVector geometry for polygon units (Queen = shared boundaries); otherwise coords columns are read from the table (Queen = lattice).

Usage

```
gw_consensus_class(
  class_dt,
  unit_col,
  geometry = NULL,
  class_col = "trend_class_05",
  coords = c("longitude", "latitude"),
  poly_id = unit_col,
  include_self = TRUE,
  max_order = 10L,
  class_levels = NULL,
  verbose = FALSE
)
```

Arguments

class_dt	A data.table/data frame stacked over settings, containing the unit id (unit_col), the categorical class (class_col), and - in point mode - the coordinate columns named in coords.
unit_col	Name of the spatial-unit identifier column (e.g. "grid_id").
geometry	Optional sf/SpatVector polygon layer (id field poly_id matching unit_col) for polygon mode. If NULL, point mode uses coords.
class_col	Name of the categorical class column. Default "trend_class_05".
coords	Length-2 character vector naming the longitude/latitude columns (point mode). Default c("longitude", "latitude").
poly_id	Name of the id field in geometry. Default: unit_col.
include_self	Logical; count the unit itself in the Queen vote. Default TRUE.
max_order	Maximum Queen order to expand to when breaking ties. Default 10.
class_levels	Optional character vector of all class labels (fixes the count columns / vote ordering). Default: sorted unique observed classes.
verbose	Logical; print progress. Default FALSE.

Value

A data.table with one row per unit: the id column, longitude, latitude, n_settings, modal_class, modal_agreement, queen_class, queen_order (order at which the vote resolved), and queen_agreement.

See Also

Other Consensus: [gw_consensus_scalar\(\)](#)

gw_consensus_scalar	<i>Consensus scalar across the kernel x distance-metric domain</i>
---------------------	--

Description

Collapses a stacked table of per-unit continuous estimates (one row per unit per distance_metric x kernel setting, within a history window) to a single per-unit consensus value via a user-supplied reducer (agg_fun, default the median), together with the spec-uncertainty spread and a sign-agreement share. Optionally reconciles each unit's consensus with its first-order Queen neighbourhood, widening the Queen order until enough neighbours are gathered. Point vs polygon mode is class-detected: pass an sf/SpatVector geometry for polygon units (Queen = shared boundaries, output coords = point-on-surface); otherwise the coords columns of value_dt are used (Queen = lattice).

Usage

```
gw_consensus_scalar(
  value_dt,
  unit_col,
  geometry = NULL,
  value_col = "estimate",
  coords = c("longitude", "latitude"),
  poly_id = unit_col,
  agg_fun = stats::median,
  probs = c(0.05, 0.95),
  queen_smooth = FALSE,
  include_self = TRUE,
  min_count = 1L,
  max_order = 10L,
  verbose = FALSE
)
```

Arguments

value_dt	A data.table/data frame stacked over settings, containing the unit id (unit_col), the continuous estimate (value_col), and - in point mode - the coordinate columns named in coords.
unit_col	Name of the spatial-unit identifier column (e.g. "grid_id").
geometry	Optional sf/SpatVector polygon layer (id field poly_id matching unit_col) for polygon mode. If NULL, point mode uses coords.
value_col	Name of the continuous value column. Default "estimate".
coords	Length-2 character vector naming the longitude/latitude columns (point mode). Default c("longitude", "latitude").
poly_id	Name of the id field in geometry. Default: unit_col.
agg_fun	Function reducing a numeric vector (the values across settings for one unit) to a single scalar. Default stats::median. Any function of the form function(x) ... works, e.g. mean, function(x) mean(x, trim = 0.1).

probs	Length-2 numeric; the lower/upper quantiles reported across settings as consensus_lo/consensus_hi. Default c(0.05, 0.95).
queen_smooth	Logical; if TRUE, also compute the Queen-neighbourhood smoothed consensus. Default FALSE.
include_self	Logical; include the unit itself in the Queen neighbourhood. Default TRUE.
min_count	Minimum number of units to gather before reducing the Queen neighbourhood; the order widens until reached. Default 1.
max_order	Maximum Queen order to expand to. Default 10.
verbose	Logical; print progress. Default FALSE.

Value

A data.table with one row per unit: the id column, longitude, latitude, n_settings, consensus, consensus_sd, consensus_mad, consensus_lo, consensus_hi, sign_agreement, and - when queen_smooth = TRUE - queen_value, queen_order, queen_agreement.

See Also

Other Consensus: [gw_consensus_class\(\)](#)

gw_distance_metric_names

List valid GW distance metric names

Description

List valid GW distance metric names

Usage

```
gw_distance_metric_names()
```

Value

Character vector of valid preset names.

gw_distance_metric_presets

GW distance metric presets for GWmodel

Description

Provides a curated set of distance metric presets (Minkowski family and great-circle) for **GWmodel**. Each preset specifies (p, theta, longlat) for `GWmodel::gw.dist()`.

Usage

```
gw_distance_metric_presets()
```

Value

A named list of presets, each entry a list(p, theta, longlat).

PACKAGE_GLOBALVARIABLES
global_names

Description

A combined dataset for global_names

Usage

PACKAGE_GLOBALVARIABLES

Format

A list of global names.

Source

Internal innovation

resolve_distance_metric
Resolve a GW distance metric preset

Description

Resolve a GW distance metric preset

Usage

resolve_distance_metric(name, stop_on_error = TRUE)

Arguments

name Character scalar. One of gw_distance_metric_names().
 stop_on_error Logical. If TRUE, throw for unknown names; else NULL.

Value

list(p, theta, longlat) or NULL.

us_state_ag_census	<i>U.S. state agricultural land and crop-insurance panel</i>
--------------------	--

Description

A state x Census-of-Agriculture-year panel of agricultural land value/area, crop-insurance coverage, and Federal Crop Insurance Program (FCIP) risk and participation measures, assembled from the USDA NASS Census of Agriculture and RMA extracts published in the USFarmSafetyNetLab data release. Bundled as example data for the geographically weighted tools in gwkrit (a real, gap-prone state-level surface for `estimate_gwr()`, `estimate_gwlag()`, `estimate_gwss()`, and the consensus tools). Join to a state polygon layer (e.g. `urbnmapr::get_urbn_map("states", sf = TRUE)`) by state FIPS.

Usage

```
us_state_ag_census
```

Format

A `data.table` with one row per (`state_code` x `census_year`), keyed on both, and the columns:

state_code State FIPS code (integer).

census_year Census of Agriculture year (integer).

crop_insurance_acres Agricultural land enrolled in crop insurance (acres).

cropland_acres Cropland (acres).

ag_land_value Total value of agricultural land and buildings (USD).

ag_land Total agricultural land (acres).

fcip_base_rate Acreage-weighted FCIP base premium rate near the census year (a proxy for assessed production risk).

fcip_acres FCIP net reported acres near the census year.

fcip_adoption FCIP participation: `pmin(fcip_acres / cropland_acres, 1)`.

ag_land_value_per_acre `ag_land_value / ag_land` (USD/acre).

Details

The example analyses use it to ask where farmland trades at a *risk discount* and whether FCIP offsets it: regress `log(ag_land_value_per_acre)` on the FCIP base rate (`fcip_base_rate`, a risk proxy), participation (`fcip_adoption`), and their interaction.

Source

USDA NASS Census of Agriculture and USDA RMA, via the piggyback release [ftsiboe/USFarmSafetyNetLab](#): `agCensusAcres.rds` and `agCensusInsurance.rds` (tag `nass_extracts`), `fcip_aph_base_rate.rds` (tag `adm_extracts`), and `sobsc_all.rds` (tag `sob`). Built by `data-raw/scripts/internal_datasets.R`.

Examples

```
data(us_state_ag_census)
us_state_ag_census[, participation := crop_insurance_acres / cropland_acres]
head(us_state_ag_census)
```

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